

Haoyang Jiang

Scientific Machine Learning | Dynamical Systems | Inverse Problems

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[Website](#) / [GitHub](#) / [Google Scholar](#) / [LinkedIn](#) / [Interactive research demos](#)

Research Profile

Ph.D. researcher developing machine-learning methods for physical dynamical systems. Research spans neural operators, physics-guided graph learning, generative inverse modeling, and data assimilation, with applications in hydrology, geophysics, and space systems.

Education

William & Mary

Ph.D. in Data Science

Expected 2027

Williamsburg, Virginia

University of Science and Technology of China

M.S. in Astrometry and Celestial Mechanics

Sep. 2020 – Jun. 2023

Hefei, China

Research: data assimilation, atmospheric-density uncertainty propagation, and numerical dynamics.

China University of Mining and Technology (Beijing)

B.S. in Geophysics; Minor in Computer Science

Sep. 2016 – Jun. 2020

Beijing, China

Research background: seismic forward modeling, inversion, and computational geophysics.

Research Experience

Physics-guided forecasting on open physical networks

Code

- Develop graph-based models for river and coastal systems, combining learned boundary forcing with physics-guided transport and long-horizon forecasting.
- Study the Danube river network and Chesapeake Bay mesh; the ICDM study reports a 38.3% average RMSE reduction relative to its spatiotemporal GNN backbone across reported horizons.

Neural operators and boundary-value problems

Research

- Develop Fredholm-integral-equation neural operators for boundary-value problems and learned ghost-node representations for missing boundary context in physical networks.
- Connect integral formulations, graph topology, and numerical propagation to improve learning when the modeled domain is incomplete.

Generative models and scientific inverse problems

FWI / Flow matching

- Combine pretrained generative priors with inference-time optimization for full-waveform inversion, without paired retraining for each new observation.
- Investigate second-order optimal-transport flow matching and variational formulations linking position and velocity in generative dynamics.

Neural differential, delay, and integral equations

Live demos

- Implement end-to-end Neural ODE, learned-delay DDE, and neural integral models for continuous-time prediction and boundary-driven sensor responses.
- Evaluate long-horizon and frequency-shift behavior with discrete and implicit adjoints, differentiable history interpolation, gradient checks, and numerical-refinement tests.

Satellite dynamics and uncertainty propagation

- Studied satellite trajectory modeling and orbital-error propagation under atmospheric-density uncertainty using numerical and machine-learning methods.

Publications

1. **H. Jiang**, Z. Wang, S. Gao, Y. J. Zhang, X. Zhu, and Y. He. *Physics-Refined Spatiotemporal Forecasting on Open-Boundary Hydrologic Graphs*. **IEEE ICDM**, 2026, accepted.
Selected among ICDM's best-ranked papers; invited to submit an extended version to KAIS.
2. **H. Jiang**, B. Qu, X. Zhu, J. Tan, and Y. He. *Boundary-Consistent Graph Neural Networks for Topological Flux Prediction*. **Transactions on Machine Learning Research**, 2026.
3. **H. Jiang** and Y. Z. Qu. *Fredholm Integral Equations Neural Operator (FIE-NO) for Boundary Value Problems*. **Machine Learning: Engineering**, 2026, accepted.
4. **H. Jiang**, J. D. Wang, Y. He, et al. *Topology-aware Neural Flux Prediction Guided by Physics*. **ICML**, 2025.
5. **H. Jiang**, M. J. Zhang, et al. *Orbital error propagation considering atmospheric density uncertainty*. **Advances in Space Research**, 71(6), 2566–2574, 2023.

Industry Experience

Origin Space | Assistant Engineer Intern

Feb. 2022 – Jun. 2022

Nanjing, China

- Applied machine-learning and numerical methods to space-technology research, including reproducing and benchmarking algorithms on engineering data.
- Improved research prototypes through data preprocessing, parameter tuning, and runtime optimization.

Technical Skills

Machine learning: Physics-guided learning, neural operators, graph neural networks, generative models, flow matching, and continuous-time neural models.

Scientific computing: ODE/PDE solvers, differentiable simulation, adjoint methods, inverse problems, data assimilation, uncertainty propagation, and numerical optimization.

Programming: Python, PyTorch, NumPy, SciPy, MATLAB, Git, and LaTeX.

Invited Talks

- *Topology-Aware Graph Neural Networks by Physics*. KGML Workshop: Leading the New Paradigm of AI for Science, University of Michigan, Ann Arbor, Aug. 2025.
- *Satellite trajectory modeling using machine learning methods*. International Symposium on Space Flight Dynamics, Beijing, Aug. 2022.
- *Orbit error propagation considering atmospheric density uncertainty*. Youth Innovation Promotion Association of the Chinese Academy of Sciences, Nanjing, Dec. 2021.

Professional Service

Reviewer: ACM Multimedia (2025); IEEE ICDM (2025, 2026); AAAI, ICLR, KDD, and NeurIPS (2026).

Honors

Outstanding Graduate Student Award, USTC, 2022 and 2023.

First-Class Scholarship, China University of Mining and Technology (Beijing), 2018 and 2019.